



COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering

ASSIGNMENT NO 4- FALL 2025

Course Title:	Operating System			Course Code:	CSC323	Credit Hours:	3(2,1)
Course	Engr. Ahmad Mudassir			Program	BCE		
Semester:	5 th	Batch:	FA23-BCE	Section:	A,B	Date:	3 rd Jan 2026
Deadline	6 th Jan 2026			Maximum Marks:	20		
Student's Name:				Reg. No.	CUI/	/LHR	

Important Instructions / Guidelines:

Note: Assignments must reflect each student's individual work. Any instance of plagiarism, duplication, or copying from other students or online sources will result in a zero (0) grade for all parties involved.

Assignment Title: DeadLocks

Description

Critically analyze the concept of deadlock in operating systems by comparing deadlock prevention, avoidance, and detection & recovery strategies.

Evaluate each approach in terms of system performance, resource utilization, flexibility, and implementation complexity.

Based on your analysis, argue and justify why most modern operating systems choose to tolerate deadlocks rather than enforce strict deadlock prevention.

Key Concepts Tested

- Thread creation and synchronization (pthread_create, pthread_join, pthread_mutex_t)
- Directory traversal (opendir(), readdir(), closedir())
- File I/O (read(), write())
- Interprocess communication (optional pipes)
- Resource sharing and protection (mutex/semaphore)

Deliverables

1. Source code files (.c)
2. Report including objective, working principle, flowchart, and output screenshots.

Test Case Table

Test Case ID	Input Directory	Description / Condition	Expected Output	Evaluation Criteria
TC1 – Basic	/testdir1	Directory with 5 files and 1 subdirectory.	Total files = 5, Total directories = 1, Total size matches actual.	Counts match; output format correct.
TC2 – Nested Recursion	/testdir_nested	Nested directories with multiple levels.	All nested files counted correctly.	Recursive traversal works; no missing files.
TC3 – Concurrency Stress	/testdir_concurrent	5 large subdirectories processed by threads.	Threads execute concurrently; total time < sequential.	Parallel execution verified; no data corruption.
TC4 – Empty / Invalid	/emptydir or /nonexistent	Empty directory or invalid path.	Prints “No files found” or error message.	Graceful handling; no crash.
TC5 – Performance	/testdir_large	Directory with 1000+ files.	Correct output within 2s (8 threads).	Improved performance; correct synchronization.

Supporting Test Setup Scripts

Use the following Bash commands to create test directories to run test cases

```
mkdir -p testdir1/sub
```

```
touch testdir1/file{1..5}.txt
```

```
touch testdir1/sub/file6.txt
```

```
mkdir -p testdir_nested/sub1/sub2/sub3
```

```
touch testdir_nested/sub1/file1.txt testdir_nested/sub1/sub2/file2.txt
```

```
mkdir -p testdir_concurrent/{A,B,C,D,E}
```

```
for d in testdir_concurrent/*; do touch $d/file{1..20}.txt; done
```

```
mkdir testdir_large
```

```
for i in {1..1000}; do echo "Hello" > testdir_large/file$i.txt; done
```

```
mkdir emptydir
```

Evaluation Criteria (Rubric)

Component	Marks	Description
Thread creation and synchronization	15	Correct usage of pthreads, mutexes
Directory traversal logic	10	Correct use of opendir/readdir
Report & Output correctness	10	Clarity, correctness, readability
Total	35	